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A PALEOMAGNETIC RESULT FROM THE VENTERSDORP LAVAS OF SOUTH AFRICA

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Samples have been collected in the vicinity of the De Beers diamond mine from at least two flows of the Ventersdorp lavas. The specimens responded equally well to both thermal and magnetic cleaning methods and a preliminary paleomagnetic pole position is presented for Ventersdorp times. The samples studied are shown not to have been influenced either magnetically or petrologically by the emplacement of the De Beers mine kimberlite in cretaceous times.

1. INTRODUCTION

In 1966 we visited Kimberley in the Cape Province of South Africa to collect samples for a paleomagnetic study of its diamond bearing pipes. In the De Beers mine (latitude 28.7°S, longitude 24.8°E) in addition to kimberlite and its associated inclusions it was found possible to sample at two levels in the country rock lavas that have been assigned to the Ventersdorp system [1] of South Africa. As no paleomagnetic measurements have been reported from rocks of the Ventersdorp system we thought it worthwhile to sample these lavas and try to establish a preliminary paleomagnetic pole position for Ventersdorp times.

2. GEOLOGICAL SETTING

The Ventersdorp system is Precambrian and is composed mainly of basic amygdaloidal lavas; sediments and acid volcanics are present but they are of local importance only. Unconformities separate the Ventersdorp system from the older Witwatersrand system and the younger Transvaal system which overlies it. Nicolaysen [2] has suggested an age of around 2100 m.y. for the system. In the Kimberley area the Transvaal and succeeding Precambrian systems are absent and the Ventersdorp system is overlain by the Dwyka series of tillites and black shales, which forms the base of the Karroo system in southern Africa. The diamond bearing pipes of the Kimberley region have pierced almost vertically the flat lying strata of the region.

3. SAMPLING AND MEASUREMENTS

The lavas were sampled on the 350-ft level (site A) where they immediately underlie the Dwyka tillite and on the 412-ft level (site B). The kimberlite pipe has roughly elliptical cross section at these depths with major and minor axes approximately 1000 ft and 500 ft. At site A nine blocks were collected, probably from two and possibly more separate flows, at distances varying from approximately 6-600 ft from the contact of the lavas with the kimberlite pipe. At site B five blocks were collected from a single flow at distances in excess of 600 ft from the contact. The blocks were oriented by means of a Brunton compass-clinometer. From every block two cores 1 in. in diameter were drilled from each

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of which a cylindrical specimen 1 in. in length was cut. The directions and intensities of magnetization of the specimens were measured using a spinner magnetometer [3]. One specimen from each block was used for alternating magnetic field demagnetization studies with the apparatus described by McElhinny [4], while the second specimen was used for thermal demagnetization experiments.

The procedure for magnetic cleaning was that outlined by McElhinny and Gough [5] and the results are recorded in table 1 together with the natural remanent magnetization (NRM) data. At both sites the directions of magnetization of the specimens were initially scattered but after magnetic cleaning good groups of directions were obtained.

Two specimens were subjected to a series of thermal demagnetization experiments. They were heated in an inert atmosphere of nitrogen to a selected temperature, cooled in field free space (± 10γ) and their remanent magnetization measured. This was repeated for successively higher temperatures up to 600°C. From 20°C up to 400°C changes in directions of magnetization were observed; between 400°C and 550°C the directions of magnetization remained constant and at 600°C it was not possible to measure the direction of remanent magnetization. The intensities of magnetization increased up to 400°C and then remained approximately constant up to 550°C; at 600°C the intensity of both specimens was considerably less than 1% of the NRM value. A suitable temperature for thermal cleaning thus appeared to be 400°C and the remaining specimens were heated to this temperature. Good groups of directions were observed for both sites and the results are shown in table 1. The initial rise in the intensity of magnetization of the two test specimens is attributed to the gradual removal of a viscous component of magnetization which approximately opposed the stable component of magnetization. A similar phenomenon was noted in several of the specimens used in the alternating magnetic field demagnetization studies. Specimens from one block from site B behaved in an anomalous manner both when subjected to the thermal and the magnetic cleaning methods. The specimens from this block were shown to be capable of acquiring a very large temporary component of magnetization when left standing for a short time in the earth's magnetic field. We feel that the specimens acquired temporary components during our demagnetization experiments and that these obscured any stable component which may have been present. The results from this block have not been included in the calculations.

The mean direction of magnetization after magnetic cleaning does not differ significantly from the mean direction observed after thermal cleaning at either site A or site B. Thus the two cleaning methods used in the Salisbury laboratories appear to have been equally effective in removing secondary components of magnetization in the lavas studied. Fig. 1 demonstrates the results of both treatments on the specimens of site A. The mean direction of magnetization of each site is recorded in table 2 giving equal weight to the results obtained by the two cleaning methods. These directions are very similar.

4. DISCUSSION

The Ventersdorp lavas in the De Beers mine were intruded by the kimberlite pipe in cretaceous times [1]. Gough and Opdyke [8] and Briden [9] have shown that the mean direction of the earth's magnetic field in cretaceous times corresponds very closely to that of the present earth's field in southern Africa, which differs

Table 1

Site	N	M_N				B_{ac} or T°C	M_{ac} or M_{TH}					Site pole	
		D	I	R	k		D	I	R	k	α_{95}	Lat.	Long.
A	9	276	+54	6.967	3.9	20-30 mwb. m^{-2}	230½	+68½	8.743	31.1	9.4°	46.3°S	18.5°W
A	9	260	+58	7.836	6.9	400°C	206½	+72½	8.874	63.7	6.5°	55.4°S	0.1°W
B	4			0.998 *		20-30 mwb. m^{-2}	205½	+69½	3.964	84.7	10.0°	59.2°S	6.0°W
B	4					400°C	197	+73	3.952	62.6	11.7°	57.9°S	8.0°E

N, number of specimens; R, length of the resultant of the N unit vectors: D, declination in degrees; I, inclination in degrees (positive downwards); M_N, NRM; M_{ac}, remanent magnetization (RM) after alternating field demagnetization in peak field B_{ac}; M_{TH}, RM after thermal demagnetization in T°C; k, estimate of Fisher's [6] κ parameter; α_{95}, semiangle of cone of 95% confidence for mean direction.

* Number refers to the 'resultant' of data which are random on Watson's [7] criterion.

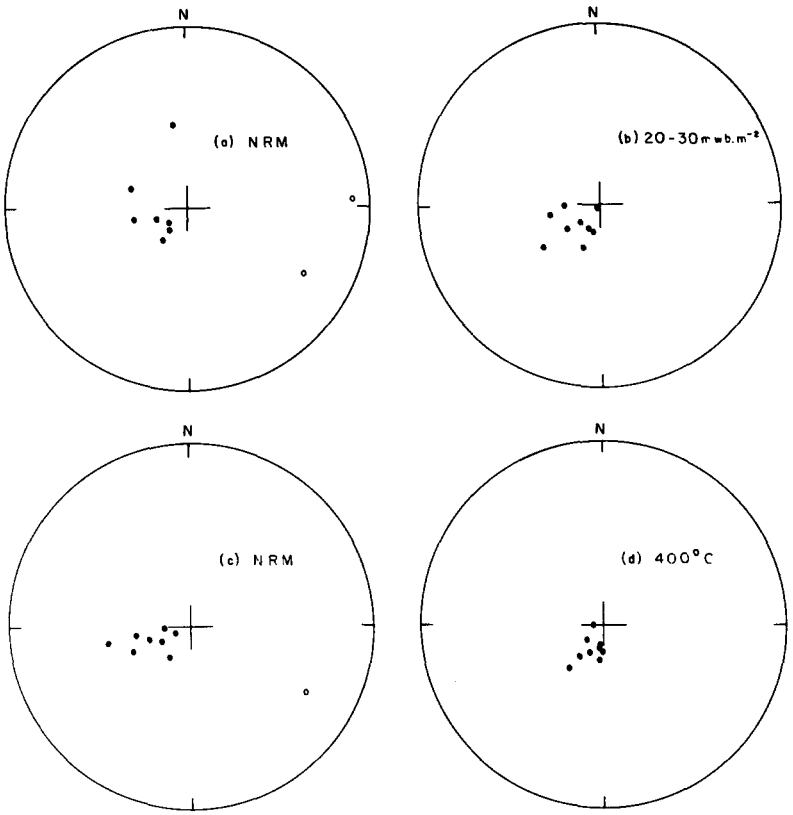


Fig. 1. Directions of magnetization of the specimens of site A. (a) and (c) show the NRM of the specimens used for magnetic and thermal cleaning, respectively. (b) and (d) respectively show the results of magnetic and thermal cleaning. Lower hemisphere stereographic projections. Solid circles, north-seeking pole down: open circles, south-seeking pole down.

markedly from an axial geocentric dipole field. Their results are confirmed by our studies [10] which indicate that the stable direction of magnetization in the De Beers kimberlite pipe is almost identical to the direction of the present earth's magnetic field at Kimberley (D = 340°, I = -65°). The mean direction of magnetization at each site in the Ventersdorp lavas differs greatly from the cretaceous direction being approximately 160° from the direction observed in the pipe.

Table 2

	D	I	Site pole	
			Lat.	Long.
Site A	219½	+71	51.1°S	10.4°W
Site B	202	+71	58.8°S	1.0°E
Sites A and B	211	+71½	55.1°S	5.2°W

D, declination; I, inclination.

Blocks from site A taken from fairly near the contact of kimberlite with the lavas exhibit stable directions of magnetization similar to those found in blocks taken more than a semi-major axis from the contact and it is not possible to distinguish the positions of the blocks with respect to the contact by their stable directions of magnetization. Studies of the variation with distance of NRM directions and intensities of magnetization both before and after demagnetization have led to the same conclusion. Thus the emplacement of the kimberlite within the pipe appears to have had no effect upon the magnetization of any of the lavas included in the present study. A study of the lavas *on* the contact and of inclusions within the kimberlite is being reported elsewhere [10].

Examination in thin sections cut from all the blocks from site A shows no difference in pe-

trology which can be correlated with their distance from the pipe. The lavas, which are andesites, have suffered alteration which probably occurred soon after extrusion.

The direction of magnetization quoted in table 2 as the mean of the two sites is thought to represent the direction of magnetization acquired by the lavas at or shortly after the time of their extrusion. Thus the pole position corresponding to this direction (table 2), calculated on the assumption of a geocentric dipole field, represents a preliminary paleomagnetic pole position relative to southern Africa for Ventersdorp times.

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